

Hi, please find below a quick walkthrough of my project and the key decisions I took during development.

1. The most interesting design/UX decision

The most interesting UX decision I made was going completely "Mobile-First" with a Bottom Navigation layout. Since the target users are contractors on a construction site, they are usually holding their phone in one hand. Placing the main actions like "Add Line Item" and "View Estimate" at the bottom of the screen makes it extremely easy for them to operate the app using just their thumb.

2. What is currently broken or fragile

To be completely transparent, the offline photo storage system is a bit fragile at the moment. Currently, I am converting high-resolution camera photos into Base64 strings and saving them directly in the browser's IndexedDB. If a user takes 50+ heavy photos for a single project, it might eventually hit the browser's storage quota or cause the app to lag. I urgently need to implement image compression before saving.

3. My creative addition and why I chose it

For my creative additions, I focused on making the app highly practical for on-site repair work. I implemented a live GPS tracker and address autocomplete feature so you can easily pin the exact project location with just one tap. To improve the overall user experience, the application automatically adapts to your device's system theme, switching smoothly between Dark and Light modes. I also added a custom comment feature for every single line item to help note down specific client instructions on the spot. Furthermore, to keep everything well documented, you can now seamlessly attach multiple photos to individual unit items, as well as upload general overview photos for the entire project.

4. What I would ship next with two more days

If I had two more days to work on this, I would definitely ship a Cloud Backup feature using Firebase or Supabase. Currently, the app works 100% offline, which is great, but if the user accidentally clears their browser data or drops their phone, their estimates are lost forever. Adding a simple sync data to the cloud would make this a fully enterprise-ready product.

Role of AI Tools in Development

AI tools played a massive role as my core pair-programming partner. I utilized AI to rapidly generate the complex UI layouts with Tailwind CSS, troubleshoot tricky Service Worker offline caching bugs, and write the heavy data-processing logic for the PDF generation and ZIP file bundling. It helped me execute my ideas and ship features in a few hours that would normally take days of manual coding and debugging.

Thank you!